

SHORT COMMUNICATION

Pretreatment with Granulocyte Colony-Stimulating Factor Reduces Myelopoiesis in Irradiated Mice

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The purpose of this study was to investigate effects of the treatment prior to irradiation with granulocyte colony-stimulating factor (G-CSF) on hematopoiesis in B10CBAF₁ mice exposed to a sublethal dose of 6.5 Gy of ⁶⁰Co γ radiation. G-CSF was administered in a 4-day regimen (3 μ g/day); irradiation followed 3 h after the last injection of G-CSF. Such a treatment was found to stimulate granulopoiesis, as shown by increased counts of granulocyte-macrophage progenitor cells (GM-CFC) and of granulocytic cells in the femoral marrow and spleen at the time of irradiation. However, postirradiation counts of GM-CFC and granulocytic cells in the marrow of mice pretreated with G-CSF were reduced up to day 18 after irradiation. Interestingly, the D_0 values for marrow GM-CFC determined 1 h after *in vivo* irradiation were 1.98 Gy for controls and 2.47 Gy for mice pretreated with G-CSF, indicating a decreased radiosensitivity of these cells after drug treatment. The inhibitory effects of the pretreatment with G-CSF on the postirradiation granulopoiesis could be attributed to the phenomenon of “rebound quiescence” which can occur after cessation of the treatment with growth factors. Postirradiation recovery of erythropoiesis in the spleen of mice pretreated with G-CSF exhibited a dramatic increase and compensated for the decreased erythropoiesis in the marrow at the time of irradiation. This complexity of the hematopoietic response should be taken into account when administering G-CSF in preirradiation regimens. © 1999 by Radiation Research Society

INTRODUCTION

The ability of various cytokines to modify the radiation response of an organism is an attractive area for study. Neta *et al.* (1, 2) have found that interleukin-1 and tumor necrosis factor, given in single injections before or shortly after irradiation, can protect mice from hematopoietic syndrome. Attention has also been focused on granulocyte colony-stimulating factor (G-CSF), a cytokine which stimulates

granulopoiesis and has already been approved for clinical use. It was demonstrated that multiple postirradiation injections of G-CSF can accelerate recovery of the hematopoietic system in mice (3, 4). However, a single preirradiation or early postirradiation injections of 1–2 μ g of G-CSF per mouse did not confer protection against hematopoietic syndrome (1, 2). A dose-dependent radioprotection in terms of survival was observed at doses of 4–80 μ g of G-CSF per mouse, with optimal results when G-CSF was administered in two doses 24 and 3 h before irradiation (5, 6). Sureda *et al.* (7) found an enhancement of postirradiation survival of mice when they injected about 20 μ g of G-CSF per mouse 2 or 24 h after irradiation. The mechanisms of these protective effects of G-CSF are not clear, but they do not appear to be due to the expansion of hematopoietic progenitor cell populations (5, 7). The actual timing, duration and dosing of G-CSF treatment in relation to radiation-induced myelosuppression appear to be critical. To further elucidate the possibilities of radioprotection conferred by G-CSF, we have investigated the action of a prolonged treatment with G-CSF before sublethal irradiation of mice. Under these conditions, an expansion of populations of bone marrow cells which respond to treatment with G-CSF can be expected (8). The daily doses of G-CSF per mouse used in this study are comparable to those employed by other authors to enhance granulopoiesis in mice under post-irradiation administration (4).

MATERIAL AND METHODS

Mice

Male B10CBAF₁ mice aged 3 months and weighing an average of 25 g were obtained from AnLab Ltd. (Prague, Czech Republic). The mice were kept under controlled conditions; standardized pelleted diet and HCl-treated tap water (pH 2–3) were available *ad libitum*. The research was conducted according to the principles enunciated in the “Guide for the Care and Use of Laboratory Animals” issued by the Czech Society for Laboratory Animal Sciences.

Irradiation

Mice were total-body-irradiated with ⁶⁰Co γ radiation at a dose rate of 0.23 Gy/min. A sublethal dose of 6.5 Gy was used for evaluation of

the postirradiation recovery of hematopoiesis. Doses of 1, 2, 3 and 4 Gy were used to determine the radiosensitivity of hematopoietic progenitor cells *in vivo*.

Drugs

Recombinant human G-CSF (Neupogen; purchased from F. Hoffman La Roche Ltd., Basel, Switzerland) was diluted with 5% glucose and injected subcutaneously at a single daily dose of 3 µg per mouse in a volume of 0.1 ml. Glucose (5%) was used for control injections.

Hematology

Blood samples were taken from the tail vein. After blood sampling, the mice were killed by cervical dislocation, the femurs and spleens were dissected, and marrow and spleen cells were harvested by standard procedures. Blood cell counts and numbers of nucleated cells of the femoral marrow and the spleen were determined using a Coulter Counter Model ZF (Coulter Electronics, UK). Differential counts were performed on smear preparations stained by the May-Grünwald-Giemsa method. The differential counts of marrow and spleen cells included the morphologically recognizable granulocytic (myeloblasts through segmented neutrophils) and erythroid (proerythroblasts through orthochromatic erythroblasts) precursor cells. The early erythroid precursors are proerythroblasts, macroblasts and basophilic erythroblasts. Bone marrow and spleen hematopoietic progenitor cells committed to granulocyte-macrophage development (granulocyte-macrophage colony-forming cells, GM-CFC) were assayed using a semisolid plasma-clot technique. Briefly, cell suspensions were plated in quadruplicate using a 10% mouse lung-conditioned medium as a source of colony-stimulating factor. Colonies (>50 cells) were counted after a 7-day incubation in a 37°C humidified environment containing 5% CO₂. In addition, endogenous hematopoietic spleen colonies were counted on day 10 after irradiation as grossly visible surface colonies on spleens fixed in Bouin's solution (nodules with a diameter of 0.5 mm or larger).

Experimental Protocol

G-CSF was administered daily from 9 to 10 a.m. for 4 days. Three hours after the last injection, the mice either underwent the hematopoietic assays (sampling denoted as "at the time of irradiation") or were irradiated. Recovery of hematopoiesis in irradiated mice was evaluated on days 10, 14, 18 and 22 after the exposure. Survival curves for GM-CFC were determined 1 h after irradiation.

Statistics

Survival curves for GM-CFC were analyzed using a program GNU-FIT (v.1.2a, C. Grammes, Riegelsberg, Germany). In all other comparisons, the Mann-Whitney rank sum test was used to determine statistical differences. The level of significance was set at $P < 0.05$.

RESULTS

Hematopoiesis in Femoral Marrow

As illustrated in Fig. 1, 4-day treatment with G-CSF induced a significant elevation of both GM-CFC counts (panel A) and granulocytic cells (panel B) at the time of irradiation together with a decrease in the count of early erythroid precursors (panel C). However, the counts of GM-CFC and of granulocytic cells were significantly decreased when compared with controls on days 10, 14 and 18 after irradiation. The counts of lymphoid cells were significantly ($P < 0.01$) decreased in mice pretreated with G-CSF at the time of irradiation; the postirradiation counts were not significantly influenced (data not shown).

Hematopoiesis in Spleen

The data given in Table 1 show that the G-CSF treatment significantly increased the counts of GM-CFC, granulocytic cells and early erythroid cells at the time of irradiation. On day 10 after irradiation, GM-CFC counts were significantly elevated in the mice pretreated with G-CSF. However, in this group a significant decrease in these cells was found on day 14 after irradiation. The treatment used did not influence the postirradiation counts of granulocytic cells significantly. On days 10 and 14 after irradiation, a significant increase in the total erythroid cells and early erythroid cells was found in mice pretreated with G-CSF. In addition, a significant increase in the number of endogenous spleen colonies was observed in this group. The preirradiation and postirradiation counts of lymphoid cells were not significantly influenced by the G-CSF pretreatment (data not shown).

Peripheral Blood Cells

As shown in Table 2, pretreatment with G-CSF significantly decreased the counts of neutrophils on day 14 and those of monocytes on day 18 after irradiation. Lymphocyte counts in the blood were significantly decreased at the time of irradiation in the group treated with G-CSF, reflecting the decrease in the number of bone marrow lymphocytes. However, a significant increase in the number of blood lymphocytes was observed on day 18 after irradiation in the mice pretreated with G-CSF. The reason for these effects is not clear. There were no distinct differences in the counts of blood erythrocytes between the groups.

Radiosensitivity of GM-CFC

The numbers of GM-CFC per femur, determined 1 h after *in vivo* irradiation with doses of 1–4 Gy, were expressed as a fraction of the preirradiation values. The experiments were repeated four times. The survival curves did not have a shoulder. The D_0 values ($\pm$ SEM) were 1.98 ± 0.09 Gy for controls and 2.47 ± 0.16 Gy for mice pretreated with G-CSF. The difference between the D_0 values is significant ($P < 0.02$) and indicates a decrease in radiosensitivity of GM-CFC after pretreatment with G-CSF.

DISCUSSION

The experiments presented in this paper show that G-CSF administered to mice in a 4-day treatment regimen enhances granulopoiesis, as indicated by increased counts of GM-CFC and granulocytic cells in the femoral marrow and spleen. However, this situation at the time of irradiation exposure does not offer any advantage for the postirradiation recovery of myelopoiesis. On the contrary, the counts of GM-CFC and of granulocytic cells in the marrow were reduced up to day 18 after irradiation. The elevated counts of GM-CFC in the spleen of mice pretreated with G-CSF

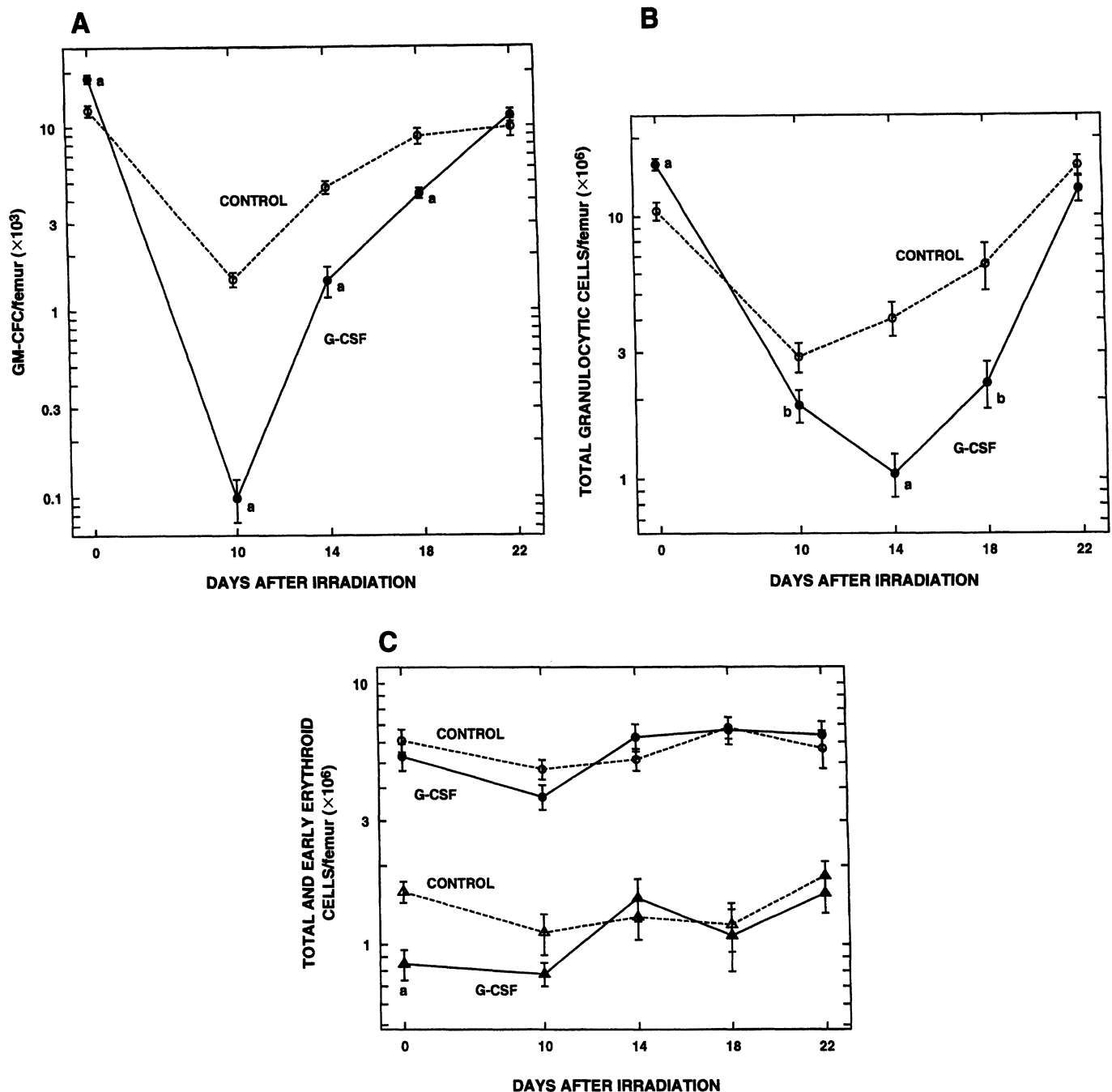


FIG. 1. Hematological indices in the femoral marrow of mice pretreated with G-CSF and control mice determined at the time of irradiation (day 0) and after irradiation with 6.5 Gy. Panel A: GM-CFC numbers; panel B: total granulocytic cells; panel C: total (circles) and early (triangles) erythroid cells. The data represent means $\pm$ SEM of values obtained from 6–10 mice. a, $P < 0.01$; b, $P < 0.05$ compared to control.

observed on day 10 after irradiation may be a consequence of the enlarged pool of these cells at the time of irradiation. The lower counts of these cells in the pretreated group on day 14, together with the lack of significant differences in the splenic granulocytic cells, indicate that the spleen could compensate for the decreased marrow granulopoiesis only in the early postirradiation period. As discussed below, the spleen of irradiated mice was repopulated mainly by erythroid cells. The inhibitory effect of pretreatment with G-CSF on myelopoiesis was reflected in lower counts of peripheral

neutrophils on day 14 and monocytes on day 18 after irradiation. The moderate expression of the effects in blood may be due to higher reserves of granulocytes under preirradiation conditions and/or to the operation of control mechanisms which are independent of the action of G-CSF. Such a role may be played by a factor inducing release of marrow granulocytes (9). Calculating the proliferative cells (myeloblasts, promyelocytes and myelocytes) and nonproliferative cells (metamyelocytes, band and segmented stages) separately, we have observed a very low level of non-

TABLE 1
Hematological Indices in the Spleen Determined at the Time of Irradiation and after Irradiation with 6.5 Gy in Control Mice and Mice Pretreated with G-CSF

			Days after irradiation	
			10	14
	At time of irradiation			
Total erythroid cells (×10 ⁶)	Control	47.12 ± 4.32	4.13 ± 0.70	46.29 ± 4.01
	G-CSF	60.66 ± 4.48	39.05 ± 9.33 ^a	86.29 ± 5.76 ^a
Early erythroid cells (×10 ⁶)	Control	6.50 ± 0.77	0.78 ± 0.16	9.29 ± 1.42
	G-CSF	14.09 ± 1.83 ^a	9.18 ± 2.28 ^a	22.42 ± 2.37 ^a
Total granulocytic cells (×10 ⁶)	Control	7.37 ± 1.38	1.10 ± 0.23	5.24 ± 0.86
	G-CSF	15.92 ± 2.29 ^a	2.69 ± 1.10	3.53 ± 0.69
GM-CFC	Control	5,697 ± 1,524	16 ± 2	1,193 ± 25
	G-CSF	50,703 ± 8,111 ^a	449 ± 133 ^a	914 ± 57 ^a
Endogenous spleen colonies	Control		6.4 ± 0.7	
	G-CSF		19.5 ± 3.2 ^b	

Notes. The data represent means $\pm$ SEM of values obtained from 5–10 mice. ^a $P < 0.01$, ^b $P < 0.05$ compared to controls.

proliferative cells in the marrow of mice pretreated with G-CSF on day 14 after irradiation (data not shown), which suggests a release of cells into blood. The ratio of nonproliferative to proliferative cells was 4.39 ± 1.73 in controls and 0.65 ± 0.26 in mice pretreated with G-CSF, the difference being significant at the 0.01 level.

Concerning the reasons for the inhibitory effects of pretreatment with G-CSF on the postirradiation granulopoiesis, increased damage to progenitor cells does not appear to play any role. On the contrary, we have found decreased radiosensitivity of GM-CFC in terms of D_0 values. This is in accord with the *in vitro* observations suggesting that the preconditioning with G-CSF renders GM-CFC less radiosensitive (10). In addition, the possibility that the action of G-CSF leads to the depletion of some early hematopoietic progenitors is unlikely after the short-term treatment regimen (4, 11). The most reasonable explanation for our findings is the mechanism of “rebound quiescence”, a feedback phenomenon which can take place after treatment with growth factor ceases (11). For example, it was shown that treatment of patients with tumors with granulocyte-macrophage colony-stimulating factor induced sustained increases in cycling of myeloid progenitors, but after administration

of the growth factor was discontinued, the proportion of S-phase marrow cells dropped to levels lower than those prior to pretreatment (12, 13). Interestingly, such a rebound effect was not observed after treatment of patients with G-CSF (14). Mechanisms like enhanced production of inhibitory cytokines, a decrease in endogenous production of the growth factors due to the exogenously induced pharmacological levels, and down-regulation of growth factor receptors on target cells could account for the effects observed in our experiments. Recently, Neta *et al.* (15) have described unpublished observations indicating that under repeated preirradiation administration interleukin-1 loses the radioprotective activity attributed to its ability to induce cycling of hematopoietic cells and even increases the radiosensitivity of mice.

The effects of G-CSF treatment on erythropoiesis and on its postirradiation behavior can be understood on the basis of earlier experimental findings. The inhibitory action of G-CSF on marrow erythropoiesis and enhancement of splenic erythropoiesis, indicated in our experiments by the changes in counts of early erythroid precursors, is a known phenomenon (16). Migration of hematopoietic progenitor cells from the marrow to the spleen during treatment with

TABLE 2
Peripheral Blood Cells (per μ l) Determined at the Time of Irradiation and after Irradiation with 6.5 Gy in Control Mice and Mice Pretreated with G-CSF

		At time of irradiation	Days after irradiation			
			10	14	18	22
Neutrophils ($\times 10^3$)	Control	1.78 $\pm$ 0.14	0.43 $\pm$ 0.03	0.39 $\pm$ 0.04	0.63 $\pm$ 0.07	1.35 $\pm$ 0.10
	G-CSF	2.60 $\pm$ 0.50	0.37 $\pm$ 0.07	0.23 $\pm$ 0.03 ^a	0.66 $\pm$ 0.11	1.68 $\pm$ 0.15
Monocytes	Control	348 $\pm$ 49	20 $\pm$ 8	49 $\pm$ 12	104 $\pm$ 19	326 $\pm$ 44
	G-CSF	569 $\pm$ 104	25 $\pm$ 5	32 $\pm$ 6	37 $\pm$ 3 ^a	322 $\pm$ 35
Lymphocytes ($\times 10^3$)	Control	4.10 $\pm$ 0.42	0.28 $\pm$ 0.04	0.66 $\pm$ 0.05	0.93 $\pm$ 0.17	0.90 $\pm$ 0.07
	G-CSF	3.00 $\pm$ 0.18 ^b	0.37 $\pm$ 0.04	0.74 $\pm$ 0.10	1.82 $\pm$ 0.29 ^a	1.20 $\pm$ 0.12
Erythrocytes ($\times 10^6$)	Control	9.53 $\pm$ 0.18	7.08 $\pm$ 0.19	6.41 $\pm$ 0.28	7.28 $\pm$ 0.31	7.51 $\pm$ 0.29
	G-CSF	9.41 $\pm$ 0.18	7.02 $\pm$ 0.15	6.61 $\pm$ 0.23	7.09 $\pm$ 0.40	7.60 $\pm$ 0.14

Notes. The data represent means $\pm$ SEM of values obtained from 9–10 mice. ^a $P < 0.01$, ^b $P < 0.05$ compared to control.

G-CSF (17, 18) may be responsible for this effect and for the enhanced formation of endogenous spleen colonies and the dramatic rise of splenic erythropoiesis on days 10 and 14 after irradiation. It is known that the spleen of rodents provides a suitable environment for erythroid differentiation and that spleen colonies are mainly of the erythropoietic type (19). The enhancement of splenic erythropoiesis can be regarded as compensation for the initial inhibition of marrow erythropoiesis. Probably for this reason, changes in erythropoiesis in the tissues were not reflected in the counts of blood erythrocytes.

To our knowledge, this is the first study showing an unfavorable effect of pretreatment with G-CSF on myelopoiesis in irradiated mice. Although it is unwise to extrapolate the effects seen in mice to effects in humans, the findings presented here should be taken into account when investigating the radioprotective ability of G-CSF as well as in clinical attempts to employ the curative activity of G-CSF in the course of radiotherapy regimens.

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